Yingru Li

szrlee@gmail.com • +65 8747 5669

Curriculum Vitae

EDUCATION

The Chinese University of Hong Kong (CUHK) *2025*

Ph.D. in Computer and Information Engineering.
Advisor: *Zhi-Quan (Tom) Luo*.
Committee: Xinyun Chen, Baoxiang Wang, John C.S. Lui, Benjamin Van Roy (Stanford & DeepMind).

Huazhong University of Science and Technology (HUST) *2020*

B.E./M.E. in Computer Science. Ranked 1st/134 overall; 1st/26 in Software and Theoretical CS.

PROFESSIONAL EXPERIENCE

xAI Singapore
Member of Technical Staff *December 2025 – Present*

- **Make core contributions to large-scale ("big-run") reinforcement learning for the Grok models.**
  - *Science* — stability, efficiency, and the core algorithms, recipes, optimizer and scaling laws; helped identify and resolve critical training instabilities that benefit every stage of the RL pipeline, contributing to one of the longest stable RL runs to date.
  - *Long Horizon* — contributed to training recipes for long-horizon agentic RL.
  - *On-policy distillation* — contributed to training recipes and research infrastructure.
  - *Infrastructure* — built and helped scale these training recipes in the large-scale distributed RL systems.

ByteDance Singapore
LLM Algorithm Research Scientist *2025 – December 2025*

- **Led research on RL training stability and efficiency for LLM reasoning and agents.**
  - Diagnosed and formalized the *RL Training Collapse* phenomenon, tracing instability to implicit off-policyness caused by the training–inference mismatch; developed Trust Region Masking (ICML'26), Optimal Token Baseline (ICML'26), Dynamic Vocabulary Pruning, and a simple LR-scheduling fix.
  - Core contributor to **veRL** (a leading open-source RL framework, ∼18k GitHub stars): authored the Rollout Correction framework for general off-policy training and contributed the Optimal Token Baseline estimator (recipes and training infrastructure).
  - Co-led project on end-to-end RL training for multi-turn tool-integrated reasoning (SimpleTIR, ICLR'26) and Triton kernel code generation agent (Dr. Kernel, ICML'26).

Tencent AI & Robotics X Shenzhen, China
Student Research Staff Member, Agent Center *2019 – 2022*

- Led research for *scalable exploration in deep RL*: HyperAgent (ICML'24) and HyperDQN (ICLR'22), and *stable off-policy learning*: Divergence-Augmented Policy Optimization (NeurIPS'19).

Microsoft Research Beijing, China
Research Intern, Theory Center *2016*

- Research on combinatorial optimization and reinforcement / bandit learning algorithms.

PUBLICATIONS

Full list at Google Scholar. ⋆ denotes equal contribution; † denotes corresponding author.

Peer-Reviewed Conference & Journal Publications

1. **The Optimal Token Baseline: Variance Reduction for Long-Horizon LLM-RL**
   Yingru Li, Jiawei Xu, Ziniu Li, Jiacai Liu, Wei Liu, Yuxuan Tong, Longtao Zheng, Zhenghai Xue, Yaxiang Zhang, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *ICML 2026.*

2. **Trust Region Masking for Long-Horizon LLM Reinforcement Learning**
   Yingru Li, Jiacai Liu, Jiawei Xu, Yuxuan Tong, Ziniu Li, Qian Liu, Baoxiang Wang. *ICML 2026.*
3. **Optimistic Thompson Sampling for No-Regret Learning in Unknown Games**
   Yingru Li, Liangqi Liu, Wenqiang Pu, Hao Liang, Zhi-Quan Luo. *IEEE Transactions on Signal Processing, 2026.*
4. **On the Scaling Flaws of Verifier-Guided Beam Search in Mathematical Reasoning**
   Fei Yu, **Yingru Li**, Benyou Wang. *Transactions on Machine Learning Research (TMLR), 2026.*
5. **Dr. Kernel: Reinforcement Learning Done Right for Triton Kernel Generations**
   Wei Liu, Jiawei Xu, **Yingru Li**, Longtao Zheng, Tianjian Li, Qian Liu, Junxian He. *ICML 2026.*
6. **Harnessing Uncertainty: Entropy-Modulated Policy Gradients for Long-Horizon LLM Agents**
   Jiawei Wang, Jiacai Liu, Yuqian Fu, **Yingru Li**, Xintao Wang, Yuan Lin, Yu Yue, Lin Zhang, Yang Wang, Ke Wang. *ICML 2026.*
7. **SimpleTIR: End-to-End Reinforcement Learning for Multi-Turn Tool-Integrated Reasoning**
   Zhenghai Xue, Longtao Zheng, Qian Liu, **Yingru Li**, Xiaosen Zheng, Zejun Ma, Bo An. *ICLR 2026.*
8. **Scalable Exploration via Ensemble++**
   Yingru Li, Jiawei Xu, Baoxiang Wang, Zhi-Quan Luo. *NeurIPS 2025.*
9. **Owl: Optimized Workforce Learning for General Multi-Agent Assistance in Real-World Task Automation**
   Mengkang Hu, Yuhang Zhou, Wendong Fan, Yuzhou Nie, Bowei Xia, Tao Sun, Ziyu Ye, Zhaoxuan Jin, **Yingru Li**, Qiguang Chen, Zeyu Zhang, Yifeng Wang, Qianshuo Ye, Bernard Ghanem, Ping Luo, Guohao Li. *NeurIPS 2025.*
10. **Q-Star Meets Scalable Posterior Sampling: Bridging Theory and Practice via HyperAgent**
    Yingru Li, Jiawei Xu, Lei Han, Zhi-Quan Luo. *ICML 2024.*
11. **Prior-Dependent Analysis of Posterior Sampling RL with Function Approximation**
    Yingru Li, Zhi-Quan Luo. *AISTATS 2024.*
12. **HyperDQN: A Randomized Exploration Method for Deep Reinforcement Learning**
    Ziniu Li, **Yingru Li**†, Yushun Zhang, Tong Zhang, Zhi-Quan Luo. *ICLR 2022.*
13. **Divergence-Augmented Policy Optimization**
    Qing Wang*, **Yingru Li***, Jiechao Xiong, Tong Zhang. *NeurIPS 2019.*

Preprints

1. **When Speed Kills Stability: Demystifying RL Collapse from the Training-Inference Mismatch**
   Jiacai Liu*, **Yingru Li***,†, Yuqian Fu, Jiawei Wang, Qian Liu, Yu Shen. *Technical Blog, 2025.*
2. **Beyond Precision: Training-Inference Mismatch is an Optimization Problem and Simple LR Scheduling Fixes It**
   Yaxiang Zhang*, **Yingru Li***,†, Jiacai Liu, Jiawei Xu, Ziniu Li, Qian Liu, Haoyuan Li. *arXiv 2026.*
3. **Learning an Opponent-Aware Anti-Jamming Strategy via Online Convex Optimization**
   Liangqi Liu, Wenqiang Pu, **Yingru Li**, Zhi-Quan Luo. *Under review (minor revision), IEEE Transactions on Signal Processing; arXiv 2026.*
4. **Taming the Tail: Stable LLM Reinforcement Learning via Dynamic Vocabulary Pruning**
   Yingru Li, Jiawei Xu, Jiacai Liu, Yuxuan Tong, Ziniu Li, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *arXiv 2025.*
5. **A Note on Hybrid Online RL and Imitation Learning for LLMs: Formulations and Algorithms**
   Yingru Li, Ziniu Li, Jiacai Liu. *arXiv 2025.*
6. **Logit Dynamics in Softmax Policy Gradient Methods**
   Yingru Li. *arXiv 2025.*
7. **Uncertainty-Aware Search and Value Models: Mitigating Search Scaling Flaws in LLMs**
   Fei Yu, **Yingru Li**, Benyou Wang. *arXiv 2025.*
8. **Probability Tools for Sequential Random Projection**
   Yingru Li. *arXiv 2024.*
9. **Simple, Unified Analysis of Johnson-Lindenstrauss with Applications**
   Yingru Li. *arXiv 2024.*

Patents

- Liangqi Liu, Wenqiang Pu, **Yingru Li**, Zhi-Quan Luo. *Game-Strategy Update Method, Device, and Medium for Unknown Matrix Games.* Chinese invention patent application, CN117787419A (Shenzhen Research Institute of Big Data; published Mar. 2024).

Research Impact & Adoption

The training–inference mismatch / RL-collapse line of work (*When Speed Kills Stability*) has been adopted across major open-source RL training stacks and frontier models:

- **DeepSeek-V3.2** adopts the geometric sequence masking to address the training–inference mismatch and off-policy training instability (arXiv:2512.02556).
- **veRL** (v0.6.0+) — core-contributed the Rollout Correction framework for general off-policy training (formulations) and the Optimal Token Baseline estimator; geometric-mean importance-sampling masking / rejection (Geo-MIS / Geo-RS) and Trust Region Masking are first-class algorithms.
- Integrated into **Slime** and **Miles**. Discussed in a **vLLM** engineering blog (bitwise-consistent on-policy RL) and a SGLang Blog.
- Used by Cognition's SWE-grep and discussed in the Qwen team technical report.

Awards & Honors

- **Best Paper Award**, 3rd Doctoral and Postdoctoral Daoyuan Academic Forum, 2024.
- **Best Student Paper**, IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM), 2024.
- **SRIBD Ph.D. Fellowship (Gold Class)**, Shenzhen Research Institute of Big Data, 2023.
- **Presidential Ph.D. Fellowship**, The Chinese University of Hong Kong, 2019–2023.
- **Tencent AI Ph.D. Fellowship**, Tencent & The Chinese University of Hong Kong, 2018.
- **National Scholarship (China)** — highest national academic honor, top 0.2% nationwide, 2018.
- **Qiming Star Award** — 1 of 5 recipients from 7,112 undergraduates, HUST, 2016.
- **First Prize**, Parallel Computation and Application Contest (PAC), Intel & CCF, 2015.
- **First Prize**, China National Mathematics Olympiad (Province-level Math League), 2012.

Peer Review & Professional Service

Reviewer for the leading peer-reviewed venues in machine learning and artificial intelligence. NeurIPS, ICML, and ICLR are the three top-tier conferences in the field; AISTATS is a leading venue in machine-learning theory and statistics; TMLR is a flagship ML journal. Complete record verifiable via OpenReview profile `~Yingru_Li1`.

Venue	Role	Papers reviewed
NeurIPS (2023, 2024, 2025, 2026)	Reviewer	6 + 6 + 5 + 5 = 22
ICLR (2024, 2025, 2026)	Reviewer	2 + 3 + 5 = 10
ICML (2025, 2026)	Reviewer†	5 + 6 = 11
AISTATS (2025)	Reviewer	2
TMLR (ongoing)	Reviewer	1
Total (2023–2026)		**46 papers**

† **ICML 2026 Gold Reviewer** — recognized by the ICML Program Chairs as among the top reviewers for the conference, based on area-chair evaluations of the quality of reviews submitted; awarded complimentary registration.

- Admitted to the AISTATS 2026 reviewer pool.
- **28 additional papers** reviewed across 9 affiliated workshops, including ICLR Workshop "Bridging the Gap Between Practice and Theory in Deep Learning," ICML Workshop "Aligning RL Experimentalists and The-

orists," and NeurIPS Workshops on Bayesian Decision-making & Uncertainty and RL4RealLife.
- **Session Chair**, 2 sessions on "Integrating Generative AI with Sequential Decision-making," INFORMS Annual Meeting 2024.
- **Organizer / Chair**, Reinforcement Learning Seminar, CUHK (8 semesters, 2019–2022).

Invited Talks & Oral Presentations

- **Revisiting Trust Region Policy Optimization in the Era of LLMs** — Invited talk, QingkeAI (Dec. 2025).
- **LLM-RL Instability and the Training–Inference Mismatch** — Invited talks at SLAI and the Bay Area RL community (Nov. 2025).
- **Tackling Data Scarcity for Trustworthy Agents** — Invited talk, ETH Zurich (Nov. 2024).
- **Exploration at Scale: Theory, Algorithms & Applications** *(a.k.a. Scalable Uncertainty Quantification for Exploration and LLM Reasoning)* — Invited talks at MIT (Jul. 2024), Princeton University (May 2024), INFORMS Annual Meeting (Seattle, Oct. 2024), International Symposium on Mathematical Programming / ISMP (Montréal, Jul. 2024), INFORMS Optimization Society Conference at Rice University (Mar. 2024), and RLChina (Jun. 2024); contributed talk at the 3rd Daoyuan Academic Forum (Jan. 2024).
- **No-Regret Learning in Unknown Games with Applications** — Invited talk, RL Theory Student Workshop, Nanjing University (Aug. 2022); contributed talk, 2nd Daoyuan Academic Forum (Aug. 2022).
- **HyperDQN: A Randomized Exploration Method for Deep RL** — Contributed talk, NeurIPS Workshop on Ecological Theory of RL (Dec. 2021).

Selected Media Coverage

- Featured for the Qiming Star Award in the HUST campus newspaper and HUST Online, 2016.

Teaching

Teaching Assistant, The Chinese University of Hong Kong (Shenzhen) — duties delivered in English (tutorials, grading, laboratory sessions): Stochastic Processes (Fall 2018), Optimization II (Spring 2019), Distributed and Parallel Computing (Fall 2019), Reinforcement Learning (Fall 2020), Matrix Analysis (Spring 2021), and Deep Learning and Their Applications (Spring 2022).